

CW4: Dissertation Thesis
**Prevalence and Genetic Risk Contribution
of the PDX1 E224K Mutation in Type 2
Diabetes Among the Sri Lankan Population.**

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**This dissertation submitted in partial fulfillment of requirements for the degree of
B.Sc. (Hons) Genetics**

Edge Hill University

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Signature:

Date:2026/03/22

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Abstract

Type 2 Diabetes Mellitus (T2DM) is a multifactorial metabolic disorder with a significant genetic component. This study investigates whether the *PDX1* E224K (c.670G>A) variant is present in a Sri Lankan population and examines its potential association with T2DM. The aim is to determine the prevalence of this variant and evaluate its contribution to disease susceptibility.

A case-control study design is used, including individuals with T2DM and normoglycaemic controls. Genomic DNA is extracted from collected samples and subjected to polymerase chain reaction (PCR) amplification targeting exon 2 of the *PDX1* gene. Amplified products are analysed using agarose gel electrophoresis and subsequently confirmed through Sanger sequencing. Statistical analyses, including Fisher's exact test and comparisons of demographic variables, are performed to assess any association between the variant and T2DM.

The results show successful amplification and sequencing in the majority of samples; however, the E224K variant is not detected in any individual. All analysed samples exhibit the homozygous wild-type genotype (GG), and no statistically significant association with T2DM is observed. Minor technical challenges, including amplification failure in a small number of samples, are noted but do not affect the overall findings.

This study concludes that the *PDX1* E224K variant is absent or extremely rare in the studied Sri Lankan population and does not contribute significantly to T2DM susceptibility. These findings highlight the importance of population-specific genetic studies and indicate the need for further research with larger sample sizes and broader genetic analysis.

Keywords

Type 2 Diabetes Mellitus (T2DM); *PDX1* gene; E224K variant; PCR; Sanger sequencing; Genetic association; Sri Lankan population

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Chapter 1: Introduction and literature review

1.1 Global and Societal Importance of Type 2 Diabetes

Type 2 diabetes mellitus (T2DM) is a major global health concern, affecting approximately 537 million adults worldwide, and is projected to rise to 783 million by 2045 (International Diabetes Federation, 2021). The burden is particularly high in low- and middle-income countries, where healthcare systems face increasing pressure from chronic disease management

(Saeedi *et al.*, 2019, Zheng *et al.*, 2018). South and East Asia have experienced a rapid increase in T2DM prevalence due to urbanisation, dietary transitions, and reduced physical activity (Hu, 2011, Chan *et al.*, 2009).

In Sri Lanka, the prevalence of T2DM is estimated to range between 10–14%, highlighting a significant and growing public health challenge (Katulanda *et al.*, 2008, Jayawardena *et al.*, 2012). The disease contributes substantially to morbidity and mortality through complications such as cardiovascular disease, nephropathy, neuropathy, and premature death.

T2DM is characterised by chronic hyperglycaemia resulting from insulin resistance and progressive pancreatic β -cell dysfunction. While environmental factors such as obesity and sedentary lifestyle are major contributors, β -cell dysfunction is increasingly recognised as a central determinant of disease progression (Prentki and Nolan, 2006, DeFronzo, 2009). This leads to impaired insulin secretion and poor glycaemic control.

Beyond its clinical impact, T2DM imposes a considerable societal and economic burden due to long-term treatment costs, reduced productivity, and increased healthcare demand. Understanding the underlying molecular mechanisms is therefore essential for improving prevention and management strategies.

1.2 Genetic Basis of T2DM and Role of Monogenic Diabetes

T2DM is a complex polygenic disorder influenced by both genetic and environmental factors. Genome-wide association studies have identified numerous susceptibility loci; however, most common variants confer only modest effects on disease risk (Voight *et al.*, 2010, Mahajan *et al.*, 2018). This suggests that rare functional variants may have a greater biological impact in specific populations.

Monogenic forms of diabetes, such as maturity-onset diabetes of the young (MODY), account for approximately 1–6% of cases and result primarily from defects in β -cell function (Fajans *et al.*, 2001, Youssef Kouidrat *et al.*, 2023). Although distinct from classical T2DM, these conditions highlight the importance of transcriptional regulation in glucose homeostasis. Emerging evidence indicates that variants in genes associated with monogenic diabetes may also contribute to early-onset or atypical T2DM, suggesting an overlap between monogenic and polygenic mechanisms (Hattersley and Patel, 2017, McCarthy, 2017).

1.3 *PDX1* as a Key Regulator of β -cell Function

PDX1 (pancreatic and duodenal homeobox 1) encodes a transcription factor essential for pancreatic development and β -cell function. During embryogenesis, *PDX1* is required for pancreatic organ formation, and complete loss-of-function mutations result in pancreatic agenesis (Jonsson *et al.*, 1994, Stoffers *et al.*, 1997).

In mature β -cells, *PDX1* regulates insulin gene transcription and controls genes involved in glucose sensing and insulin secretion (Ohlsson *et al.*, 1993, Ahlgren *et al.*, 1998). Reduced *PDX1* activity impairs insulin biosynthesis and disrupts glucose-stimulated insulin secretion. In addition to genetic mutations, epigenetic modifications such as DNA methylation have been shown to reduce *PDX1* expression in T2DM, contributing to β -cell dysfunction (Yang *et al.*, 2012, Liu *et al.*, 2021). Despite its critical role, studies investigating *PDX1* variants in T2DM remain limited and often population-specific.

1.4 *PDX1* Variants and Their Contribution to T2DM

Recent studies have identified rare *PDX1* variants in individuals with early-onset T2DM, suggesting that partial impairment of transcription factor function may contribute to disease susceptibility (Flannick and Florez, 2016, Lian *et al.*, 2023). These findings support the role of rare variants with larger functional effects in complex diseases.

However, existing studies have several limitations. Many are conducted in specific populations, limiting generalisability to other ethnic groups. Sample sizes are often small, and replication studies are limited. Furthermore, there is a lack of data from South Asian populations, including Sri Lanka, despite the high regional burden of T2DM.

These limitations highlight the need to investigate specific functionally significant variants of *PDX1* in underrepresented populations.

1.5 Functional Significance of the E224K Variant

The E224K (c.670G>A; p.Glu224Lys) variant is a missense mutation located within the highly conserved homeodomain of *PDX1*, a region critical for DNA binding and transcriptional regulation. Structural alterations in this domain can significantly impair protein function. Functional studies have demonstrated that the E224K substitution reduces insulin promoter transactivation, indicating a direct effect on β -cell transcriptional activity (Cockburn *et al.*, 2004). The replacement of a negatively charged glutamic acid with a positively charged lysine alters DNA-binding affinity and disrupts transcriptional regulation.

This variant was first identified in an Indo-Trinidadian family with diabetes, supporting its pathogenic potential (Cockburn *et al.*, 2004). Subsequent studies have reported *PDX1* variants in individuals with early-onset diabetes (Lian *et al.*, 2023, Yin *et al.*, 2024). However, these findings remain population-specific and lack representation from South Asian populations. The E224K variant was specifically selected for investigation because it is located within the highly conserved *PDX1* homeodomain. Mutations in this region are known to have significant functional consequences. Unlike many variants with uncertain clinical relevance, E224K has been experimentally shown to impair insulin promoter activation, directly impacting β -cell function. Furthermore, its identification in populations with shared ancestry suggests potential relevance to the Sri Lankan population. Despite this, its prevalence and contribution to T2DM have not been investigated in this context, making it a strong candidate for population-specific genetic analysis.

1.6 Research Gap and Justification

Despite the increasing prevalence of T2DM in Sri Lanka (Sri Lanka) (Katulanda *et al.*, 2008, Jayawardena *et al.*, 2012), molecular genetic studies investigating rare diabetes-associated variants remain limited. Most local studies have focused on epidemiological and clinical aspects, with minimal emphasis on genetic characterisation (Wijesuriya *et al.*, 2019). Existing research on *PDX1* variants is largely restricted to specific populations and often limited by small sample sizes, reducing the generalisability of findings (Flannick & Florez, 2016, Lian *et al.*, 2023). Notably, there is no available data on the presence or impact of the E224K variant in the Sri Lankan population.

Given the functional significance of this mutation (Cockburn *et al.*, 2004) and the high burden of T2DM in the region, investigating its prevalence addresses a clear gap in current knowledge. This study is therefore justified both scientifically and in a population-specific context.

1.7 Novelty and Contribution of the Study

This study represents one of the first investigations into the presence of the *PDX1* E224K variant in the Sri Lankan population. By focusing on a functionally significant mutation within a key transcription factor, this research contributes to understanding the genetic basis of β -cell dysfunction in T2DM (Ohlsson *et al.*, 1993, Cockburn *et al.*, 2004).

The findings may improve disease classification by identifying individuals with potential monogenic contributions to diabetes (Hattersley & Patel, 2017). In addition, this study supports

the development of precision medicine approaches through improved diagnostic accuracy and personalised treatment strategies (Flannick & Florez, 2016).

Furthermore, this research contributes to expanding genetic research in Sri Lanka and provides baseline data for future large-scale genomic studies (Mahajan et al., 2018, Lian et al., 2023).

1.8 Hypothesis, Aim, Research Objectives and Research Questions

Hypothesis

The *PDX1* E224K variant is present within the Sri Lankan population and occurs at a higher frequency among individuals diagnosed with T2DM compared to normoglycaemic controls, suggesting a potential contribution to disease susceptibility.

Aim

To determine the prevalence of the *PDX1* E224K mutation and evaluate its association with T2DM in Sri Lanka.

General Objective

To investigate the genetic contribution of the *PDX1* E224K variant to T2DM in the Sri Lankan population.

Specific Objectives

1. To amplify and sequence the region of the *PDX1* gene encompassing the E224K variant.
2. To determine the frequency of the mutation in individuals with T2DM and healthy controls.
3. To compare mutation frequencies between groups using appropriate statistical analysis.
4. To evaluate the association between the E224K variant and T2DM.

Research Questions

1. What is the prevalence of the E224K variant in the study population?
2. Is there a statistically significant association between the E224K mutation and T2DM?

1.9 Organization of the Thesis

This thesis is structured into four chapters. Chapter 1 provides the background and literature review on Type 2 Diabetes Mellitus, the role of *PDX1* in β -cell function, and the functional

significance of the E224K variant, including the research gap, study rationale, aims, and objectives. Chapter 2 details the materials and methods employed in the study, including the study design, participant selection, laboratory procedures such as DNA extraction, PCR amplification, sequencing, and the statistical methods used to assess the association between the E224K variant and T2DM. Chapter 3 presents the results, including the prevalence of the E224K variant, allele and genotype frequencies, and statistical analyses comparing individuals with T2DM and normoglycaemic controls. Finally, Chapter 4 discusses the findings in the context of existing literature, highlights clinical and genetic relevance, addresses study limitations, and provides recommendations for future research and public health applications.

Chapter 2: Materials and Methods

2.1 Study Design and Sampling Strategy

A pilot case–control study was conducted to evaluate the frequency of the *PDX1* E224K (c.670G>A) variant and its potential association with Type 2 diabetes mellitus (T2DM) in a Sri Lankan cohort.

A total of 12 genomic DNA samples were obtained from a clinical genetic biobank. The case group comprised six individuals clinically diagnosed with T2DM, while the control group included six normoglycaemic individuals with no documented history of diabetes.

All samples were originally derived from peripheral blood (5 mL) collected in EDTA tubes to prevent coagulation and preserve DNA integrity for downstream molecular analyses. Due to

limited sample availability, a formal sample size calculation was not performed. Therefore, this study was designed as a pilot investigation to assess feasibility and generate preliminary data. However, the small sample size substantially limits statistical power, and findings should be interpreted as exploratory rather than confirmatory. For adequately powered genetic association studies, sample size calculations should consider allele frequency, expected effect size (odds ratio), statistical power (typically 80%), and significance level ($\alpha = 0.05$) (Charan & Biswas, 2013).

All laboratory procedures were carried out at Genelabs Medical Laboratory, Sri Lanka.

2.2 Sample Size Considerations and Interpretation

Allele frequency data from South Asian populations indicate that the *PDX1* E224K variant occurs at approximately 1% (0.01) (ClinVar, NCBI; Accession: VCV000008863.23). Based on conservative assumptions, the sample size required to detect an association with 80% statistical power ranges from approximately 97 participants per group for a modest effect ($p_1 = 0.10$ in cases; $p_2 = 0.01$ in controls) to over 2,300 participants per group for a small effect ($p_1 = 0.02$; $p_2 = 0.01$). Detailed sample size calculations are provided in Appendix A.

These estimates highlight the large sample sizes required to detect modest genetic effects, particularly for variants with low penetrance. The present study, comprising six cases and six controls, is therefore substantially underpowered for definitive association testing. Accordingly, the results should be interpreted as preliminary and used to inform the design of future large-scale genetic studies in Sri Lankan populations.

2.3 DNA Samples and Quality Assessment

Pre-extracted human genomic DNA samples were obtained from the Genelabs Medical genetic biobank. DNA had been previously isolated from peripheral whole blood using a silica columnbased automated extraction system (TAN Bead Maelstrom™ TM4800, Taiwan) according to the manufacturer's protocol.

Samples were anonymised prior to analysis, and no identifiable patient information was accessible. DNA samples were stored at -20°C until use. Prior to PCR amplification, samples were vortexed and briefly centrifuged to ensure homogeneity.

DNA concentrations ranged from 2.65 to 211.35 ng/ μL . Purity was assessed using a NanoDrop spectrophotometer (Thermo Fisher Scientific, USA), with A260/A280 ratios ranging from 1.53 to 3.33 and A260/A230 ratios ranging from 0.03 to 1.91. Although some samples showed elevated purity ratios suggestive of potential contaminants, all samples yielded successful

amplification. DNA integrity was further confirmed by agarose gel electrophoresis, demonstrating intact genomic DNA suitable for downstream applications.

Tested Gene	Sample ID	DNA Concentration (ng/μL)	A260/A280	A260/A230
<i>PDX1</i>	CC 016	107.95	2.73	1.04
	CC 018	2.65	0	0.63
	CC 020	211.35	2.30	1.23
	CC 022	91.10	2.57	1.00
	CC 023	96.85	2.48	0.21
	CC 024	174.90	2.56	1.24
	DM 017	47.75	1.77	0.51
	DM 019	65.75	1.85	0.03
	DM 020	65.70	2.46	0.53
	DM 021	48.95	3.33	1.91
	DM 023	107.15	2.28	0.66
	DM 009	53.60	1.53	0.49

Table 2.1: DNA concentration and purity ratios of *PDX1* gene samples from healthy controls (CC) and Type 2 Diabetes Mellitus patients (DM). DNA Concentration is in ng/μL, and A260/A280 and A260/A230 indicate DNA purity.

2.4 PCR Amplification of the *PDX1* Target Region

2.4.1 Primer Design

The exon 2 region of the *PDX1* gene encompassing the E224K (c.670G>A) mutation site was amplified using gene-specific primers. Primers were designed and provided by Genelabs Medical Laboratory based on the reference *PDX1* gene sequence (NCBI Reference Sequence: NM_000209), ensuring coverage of the mutation site and compatibility with bidirectional Sanger sequencing.

Primer	Sequence (5'–3')	T _m (°C)	GC (%)	Target Region	Product Size
<i>PDX1</i> –E224K–F2	5'–GGAGGAGAACAAAGCGGGACG–3'	64.3	63	Exon 2	450 bp

<i>PDX1</i> – E224K–R2	5'–CCTCTCATCGTGGTTCCTGC–3'	63.9	60	Exon 2	450 bp
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Table 2.2: Primers used for amplification of the *PDX1* exon 2 region. T_m: melting temperature of the primer (°C); GC (%): percentage of guanine (G) and cytosine (C) nucleotides in the primer sequence.

Product size: expected PCR amplicon length.

Primers were synthesised commercially and diluted to a working concentration of 10 µM using nuclease-free water. Further details on primer synthesis, quality parameters, and specifications are provided in Appendix B.

2.4.2 PCR Reaction Mixture

PCR amplification was performed in a 16 µL (1 Sample) reaction mixture as summarised below:

Component	Volume (µL)
5× PCR Master Mix	3.0
Forward Primer (0.8 µM)	0.8
Reverse Primer (0.8 µM)	0.8
Genomic DNA Sample	2.0
DMSO	1.0
Nuclease-free water	8.4
Total Volume	16.0

Table 2.3: PCR reaction mixture for amplification of the *PDX1* exon 2 region including DMSO for 1 Sample.

2.4.3 PCR Cycling Conditions

Amplification was carried out using a Veriti™ Thermal Cycler (Applied Biosystems, USA) under the following conditions: initial denaturation at 95°C for 3 minutes, followed by 35 cycles of denaturation at 95°C for 30 seconds, annealing at 60°C for 30 seconds, and extension at 72°C for 1 minute, with a final extension at 72°C for 5 minutes. and a hold at 4°C until further use.

Negative controls were included to monitor contamination, and all reactions were performed in duplicate.

2.5 Agarose Gel Electrophoresis

PCR products were analysed using 2.0% (w/v) agarose gel electrophoresis prepared in 1× TAE buffer. Ethidium bromide (0.6 µg/mL) was used for DNA visualisation under UV illumination. A volume of 3.5 µL of each PCR product was mixed with loading dye and loaded into the gel wells. A 100 bp DNA ladder (Thermo Fisher Scientific, USA) loaded at 4 µL and it was used as a molecular size marker. Electrophoresis was performed at 70 V for 15 minutes. DNA bands were visualised using a Gel Doc™ XR+ imaging system (Bio-Rad, USA).

2.6 PCR Product Purification (Bead Purification)

PCR products were purified using a magnetic bead-based PCR clean-up system according to the manufacturer's instructions. Briefly, PCR products were incubated with magnetic beads, washed with 80% ethanol, and eluted in 25 µL nuclease-free water. Purified DNA was stored at −20°C until sequencing.

2.7 Sanger Sequencing

2.7.1 Sequencing Reaction

Purified PCR products were subjected to Sanger sequencing using the BigDye® Terminator v3.1 Cycle Sequencing Kit (Applied Biosystems, USA). Sequencing reactions were prepared in a total volume of 10 µL according to the manufacturer's protocol.

2.7.2 Cycle Sequencing Conditions

Cycle sequencing was performed under the following conditions: initial denaturation at 96°C for 1 minute; 25 cycles of denaturation at 96°C for 10 seconds, annealing at 50°C for 5 seconds, and extension at 60°C for 4 minutes; followed by a hold at 4°C. Negative controls (no template) were included in each run to monitor contamination, and all reactions were performed in duplicate.

2.7.3 Post-Sequencing Purification

Residual dye terminators were removed using an EDTA/ethanol precipitation method. Briefly, 5 µL of EDTA was added to each reaction, followed by 60 µL of 100% ethanol. Samples were

incubated at room temperature for 15 minutes, centrifuged to pellet DNA, washed with 80% ethanol, air-dried, and resuspended in 10 µL of nuclease-free water.

2.7.4 Capillary Electrophoresis and Sequence Analysis

Sequencing products were analysed using a SeqStudio™ Genetic Analyzer (Applied Biosystems, USA- Instrument Serial No.232002095). Bidirectional sequencing was performed for all samples using both forward and reverse primers to ensure accurate variant detection. Raw chromatogram files (.ab1) were analysed using BioEdit version 7.0.9. Forward and reverse sequences were aligned and manually edited prior to analysis. Sequence alignment was performed against the reference *PDX1* gene sequence (NCBI: NM_000209) using ClustalW. Variant identification was conducted through direct comparison with the reference sequence. All chromatograms were independently reviewed to confirm variant calls and exclude sequencing artefacts.

All laboratory procedures were conducted following institutional safety protocols. A complete risk assessment is provided in Appendix C.

2.8 Statistical Analysis

Variant frequencies were calculated separately for T2DM and control groups. Fisher's exact test was used to compare categorical variables (presence versus absence of the variant) due to the small sample size. Statistical significance was defined as $p < 0.05$.

Given the pilot nature of the study, statistical analysis was considered exploratory, and results were interpreted cautiously rather than as definitive evidence of association. Analyses were performed using IBM SPSS Statistics version 27 (IBM Corp., Armonk, NY, USA).

2.9 Ethical Approval

Ethical approval for this study was obtained from the institutional ethics committee (Reference: AAJ/ETH/COM/2024). The study was conducted in accordance with the principles of the Declaration of Helsinki.

All DNA samples were anonymised prior to analysis, and no identifiable participant information was accessible. Due to the use of archived samples and minimal associated risk, the requirement for individual informed consent was waived. The ethical clearance document is provided in Appendix D, Also blood collection consent form document is provided in Appendix E.

2.10 Data Analysis Pipeline

The experimental workflow followed a sequential pipeline:

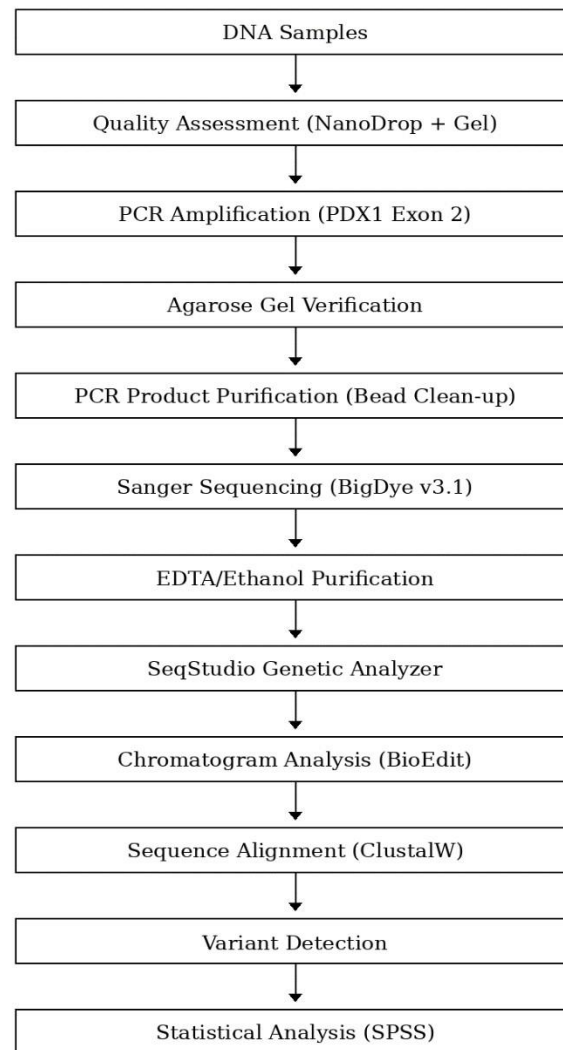


Figure 1: Workflow of PCR Amplification, Sanger Sequencing, and Variant Analysis for Detection of the PDX1 E224K Variant

Chapter 3: Results and Analysis

3.1 PCR Amplification of the *PDX1* Gene

PCR amplification of the *PDX1* exon 2 region was successfully achieved for 10 out of 12 samples. Agarose gel electrophoresis showed clear and distinct DNA bands at approximately 450 base pairs, corresponding to the expected amplicon size.

All samples demonstrated consistent amplification with no observable non-specific bands. PCR reactions were performed in duplicate, and identical banding patterns were observed across replicates.

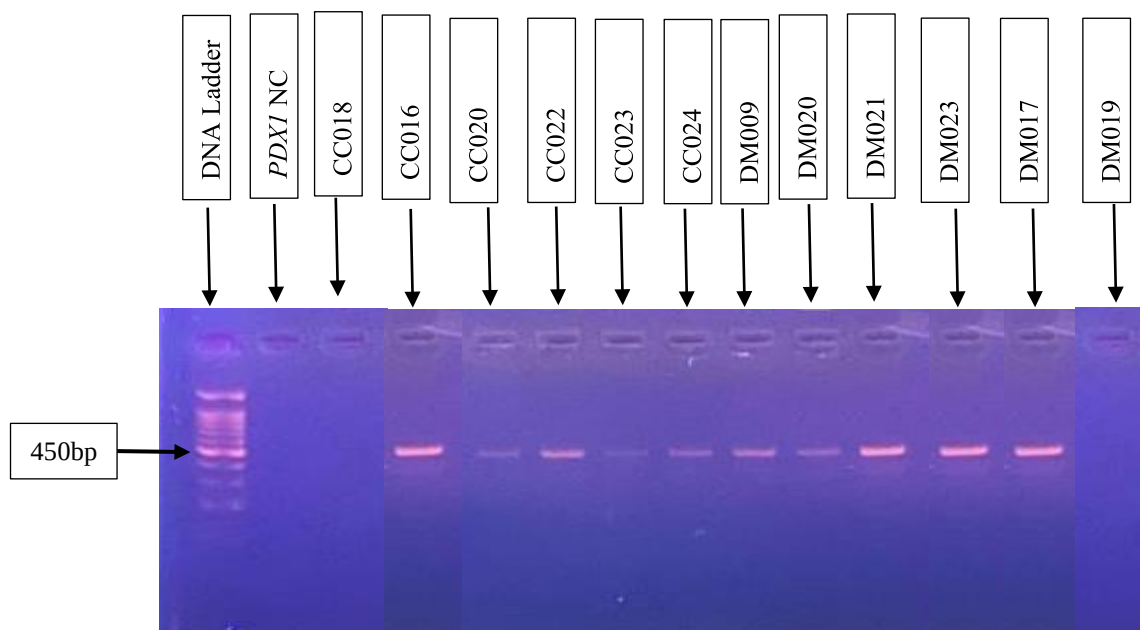


Figure 2: Agarose gel electrophoresis of PCR-amplified *PDX1* exon 2 region

PCR products were separated on a 2.0% agarose gel. Lane M (molecular marker) represents the 100 bp DNA ladder, and NC represents the negative control. Lanes correspond to individual samples from control (CC) and T2DM (DM) groups, including CC016, CC020, CC022, CC023, CC024, DM009, DM017, DM020, DM021, and DM023.

Clear bands at approximately 450 bp were observed in 10 samples, confirming successful amplification of the target region. Two samples did not show amplification. That was CC018 and DM019. No bands were observed in the negative control.

3.2 Sanger Sequencing and Identification of the E224K Mutation Site

Sanger sequencing was performed to analyse the nucleotide at position c.670 of the *PDX1* gene. High-quality chromatograms were obtained for all samples.

A clear single peak corresponding to guanine (G) was observed at position c.670 in all successfully sequenced samples (n = 10). No secondary peaks or base substitutions were detected, indicating the absence of the E224K (c.670G>A) mutation. Additional sequence data for other 9 samples are provided in Appendix F.

Sample ID: CC016

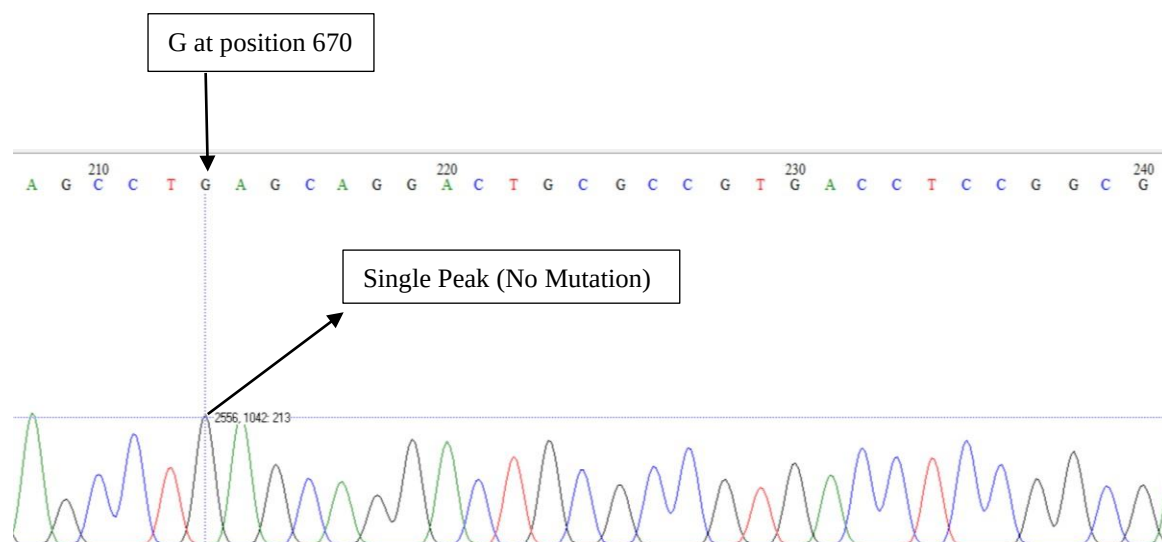


Figure 3: Representative Sanger sequencing chromatogram of the *PDX1* gene, while similar results were obtained for all successfully sequenced samples (n = 10)

The chromatogram shows a clear single peak corresponding to guanine (G) at position c.670. No secondary peaks or base substitutions were observed, indicating absence of the E224K (c.670G>A) mutation.

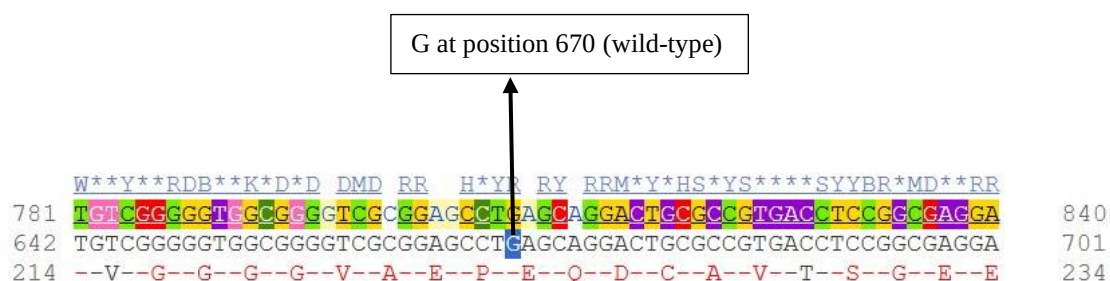


Figure 4: Reference sequence of the *PDX1* gene showing the E224K mutation site

A reference sequence of the *PDX1* exon 2 region obtained from the Ensembl database is shown, with the nucleotide at position c.670 highlighted. The reference nucleotide at this position is guanine (G), corresponding to the wild-type allele. This position represents the site of the E224K (c.670G>A) mutation.

3.3 Prevalence of the *PDX1* E224K Variant

The *PDX1* E224K (c.670G>A) variant was not detected in any of the analysed samples.

All individuals were homozygous for the wild-type allele (GG) at position c.670. No heterozygous (GA) or homozygous mutant (AA) genotypes were identified in either the T2DM or control groups.

The mutation frequency was 0% in both groups.

Group	Mutation Present	Mutation Absent	Total
T2DM	0	6	6
Control	0	6	6
Total	0	12	12

Table 3.1: Frequency of the *PDX1* E224K variant in study groups

The results showed that all analysed individuals carried the wild-type allele at the E224K position.

3.4 Description of Study Population

A total of 12 samples were included in the analysis, consisting of 6 individuals diagnosed with Type 2 Diabetes Mellitus (T2DM) and 6 normoglycaemic controls.

The mean age differed between the T2DM group (61.67 ± 13.31 years) and the control group (56.00 ± 7.10 years). The distribution of gender varied between groups, with a higher proportion of males observed in both the T2DM (83.3%) and control (66.7%) groups.

The mean Body Mass Index (BMI) also differed between the T2DM group (24.42 ± 5.08 kg/m²) and the control group (27.68 ± 8.75 kg/m²). Additional data and detailed results are presented in Appendix G.

Variable	T2DM (n=6)	Control (n=6)
Age (years, mean $\pm$ SD)	61.67 ± 13.31	56.00 ± 7.10
BMI (kg/m ² , mean $\pm$ SD)	24.42 ± 5.08	27.68 ± 8.75
Male (%)	83.3%	66.7%
Female (%)	16.7%	33.3%

Table 3.2: Demographic characteristics of study participants

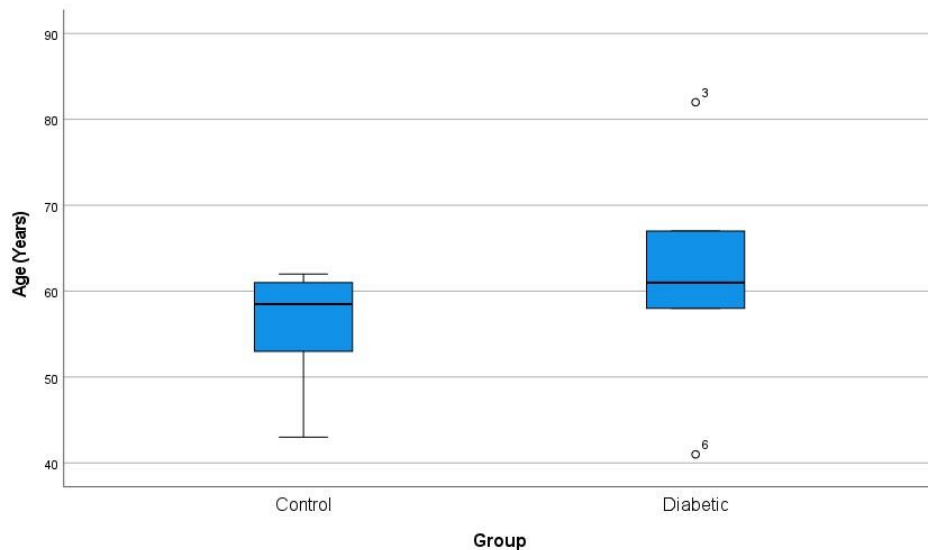


Figure 5: Distribution of Age Among Control and T2DM Groups :This box plot shows the distribution of age in the control and T2DM groups. The T2DM group demonstrates a wider range and presence of outliers, indicating greater variability in age compared to the control group.

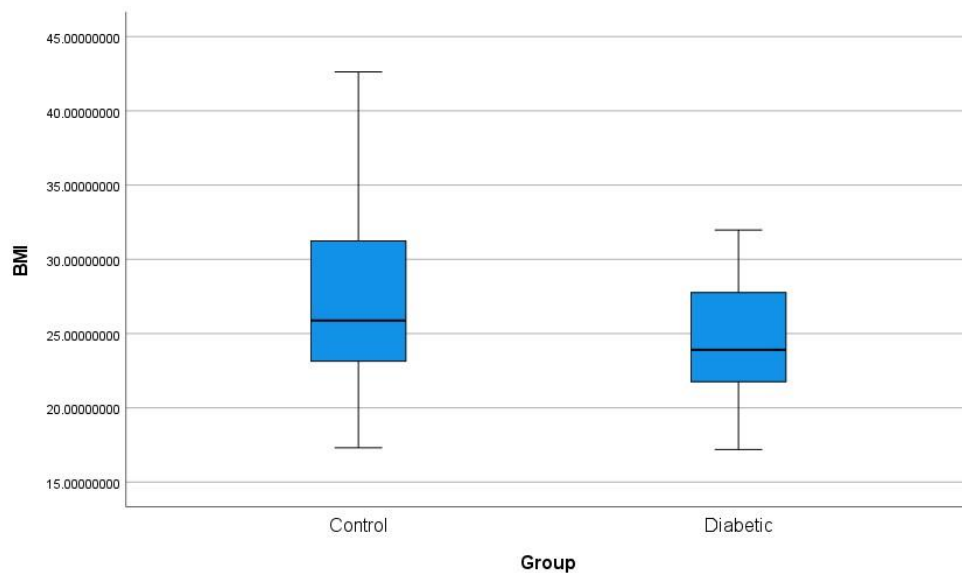


Figure 6: Distribution of Body Mass Index (BMI) Among Control and T2DM Groups :This box plot illustrates the distribution of BMI in both groups. The control group exhibits a wider spread and higher variability, while the T2DM group shows a more concentrated distribution of BMI values.

3.5 Duration of Diabetes

Descriptive statistical analysis was used to summarize the duration of diabetes among T2DM participants.

The duration of diabetes ranged from 3 to 20 years, with a mean duration of 10.0 ± 6.87 years.

Descriptive Statistics					
	N	Minimum	Maximum	Mean	Std. Deviation
Diabetic_Duration (Years)	6	3	20	10.00	6.870

Table 3.3: Descriptive statistics of diabetes duration among T2DM participants

3.6 Statistical Analysis

Statistical analyses were performed using SPSS to evaluate differences between T2DM patients and control individuals, and to assess the association between the *PDX1* E224K variant and T2DM.

3.6.1 Association Between *PDX1* E224K Variant and T2DM

Fisher's exact test was used to assess the association between the *PDX1* E224K (c.670G>A) variant and T2DM.

The variant was not detected in any of the analysed samples (n=10) in either group. 2 samples were not included due to unsuccessful amplification. Therefore, no statistically significant association was observed between the *PDX1* E224K variant and T2DM ($p > 0.05$). Additional data and detailed results are presented in Appendix H.

Group * Mutation_Present Crosstabulation						
			Mutation_Present			Total
				N/A	No	
Group	Control	Count	0	1	5	6
		% within Group	0.0%	16.7%	83.3%	100.0%
		% within Mutation_Present	0.0%	50.0%	50.0%	46.2%
	Diabetic	Count	0	1	5	6
		% within Group	0.0%	16.7%	83.3%	100.0%
		% within Mutation_Present	0.0%	50.0%	50.0%	46.2%
Total		Count	1	2	10	13
		% within Group	7.7%	15.4%	76.9%	100.0%
		% within Mutation_Present	100.0%	100.0%	100.0%	100.0%

Table 3.4: Association Between *PDX1* E224K Variant and T2DM using Fisher's exact test. No -No mutation; N/A- Not available data

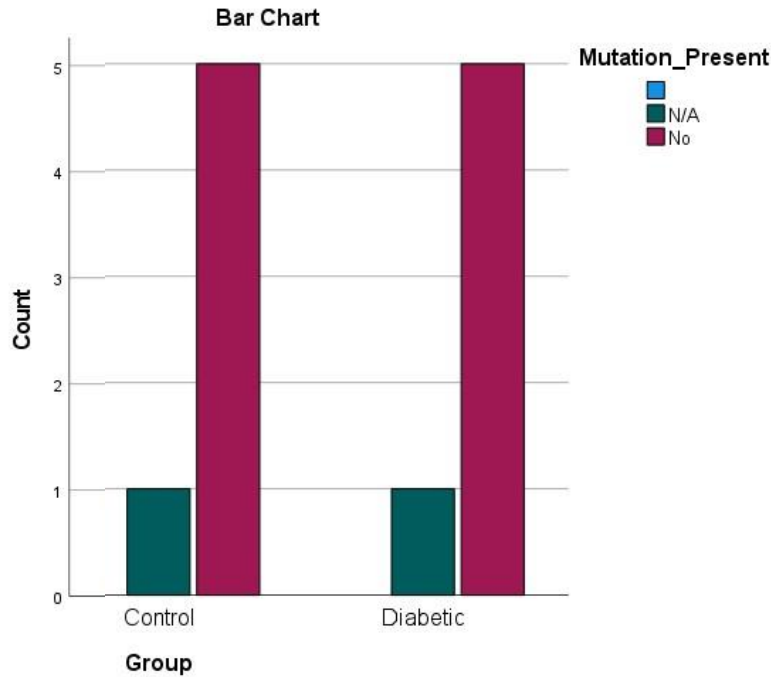


Figure 7: Comparison of Mutation Presence Between Control and Diabetic Groups Using Fisher's Exact Test

This bar chart illustrates the distribution of mutation presence among control and diabetic groups. The majority of individuals in both groups showed no detected mutation, while a very small number of samples ($n=2$) were categorized as not available (N/A). No cases with confirmed mutation were observed in either group.

The results of Fisher's Exact Test indicate that there is no statistically significant association between group type (control vs. diabetic) and mutation presence. This suggests that the studied mutation does not differ between the two groups in this dataset

3.6.2 Comparison of Age Between Groups

An independent samples t-test was conducted to compare the mean age between the T2DM and control groups.

The mean age of the T2DM group (61.67 ± 13.31 years) was higher than that of the control group (56.00 ± 7.10 years). However, this difference was not statistically significant ($p > 0.05$). Additional data and detailed results are presented in Appendix I.

Group Statistics

	Group	N	Mean	Std. Deviation	Std. Error Mean
Age (Years)	Diabetic	6	61.67	13.307	5.432
	Control	6	56.00	7.099	2.898

Table 3.5: Comparison of Age Between Groups using independent samples t-test

3.6.3 Comparison of BMI Between Groups

An independent samples t-test was performed to compare Body Mass Index (BMI) between the two groups.

The mean BMI of the T2DM group ($24.42 \pm 5.08 \text{ kg/m}^2$) differed from that of the control group ($27.68 \pm 8.75 \text{ kg/m}^2$). This difference was not statistically significant ($p > 0.05$). Additional data and detailed results are presented in Appendix J.

Group Statistics					
	Group	N	Mean	Std. Deviation	Std. Error Mean
BMI	Diabetic	6	24.4176822	5.07658047	2.07250530022
	Control	6	27.6799433	8.74607637	3.57057072923

Table 3.6: Comparison of BMI Between Groups using independent samples t-test

3.6.4 Gender Distribution Analysis

Chi-square test was used to analyse the distribution of gender between the T2DM and control groups. Due to the small sample size, Fisher's exact test was applied where appropriate. Although variation in gender distribution was observed, the difference was not statistically significant ($p > 0.05$). Additional data and detailed results are presented in Appendix K.

Group * Gender Crosstabulation

			Gender			Total
				Female	Male	
Group						
	Control	Count	0	4	2	6
		% within Group	0.0%	66.7%	33.3%	100.0%
	Diabetic	Count	0	5	1	6
		% within Group	0.0%	83.3%	16.7%	100.0%
Total		Count	1	9	3	13
		% within Group	7.7%	69.2%	23.1%	100.0%

Table 3.7: Gender Distribution Analysis using chi-square test

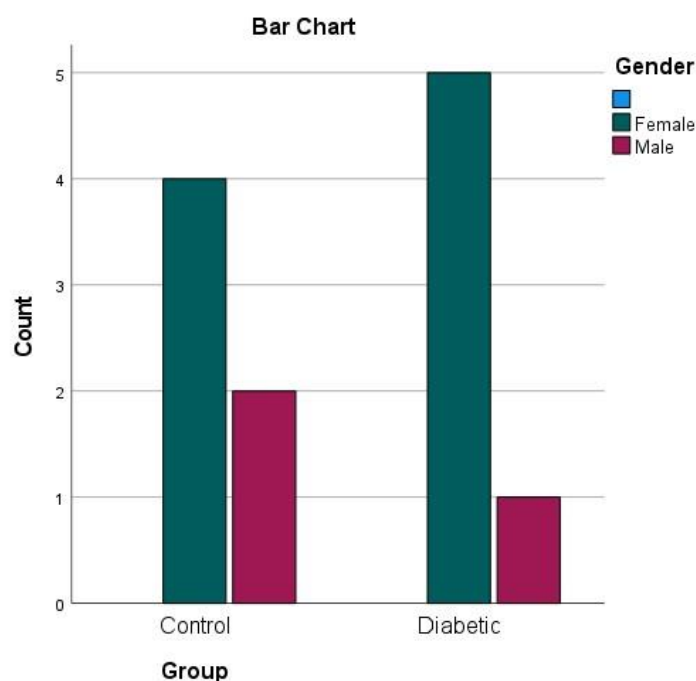


Figure 8: Gender Distribution Between Control and T2DM Groups :This bar chart illustrates the distribution of gender among the control and Type 2 Diabetes Mellitus (T2DM) groups. In the control group, females (n = 4) were more prevalent than males (n = 2). Similarly, in the T2DM group, females (n = 5) outnumbered males (n = 1).

Chapter 4: Discussion

This study aimed to determine the prevalence of the *PDX1* E224K (c.670G>A) variant and evaluate its potential association with Type 2 Diabetes Mellitus (T2DM) in a Sri Lankan population. The working hypothesis proposed that this variant would be present and occur at a higher frequency in individuals with T2DM compared to normoglycaemic controls. However, the findings of this study do not support this hypothesis. The E224K variant was not detected in any of the analysed samples, with all individuals exhibiting the homozygous wild-type

genotype (GG). Consequently, no statistically significant association between this variant and T2DM was observed.

In relation to the primary aim, the study successfully determined that the prevalence of the E224K variant in this cohort is 0%. This directly addresses the first research question, indicating that the mutation is either absent or extremely rare in the studied Sri Lankan population. Similarly, the second research question regarding association was addressed through Fisher's exact test, which demonstrated no statistically significant relationship between the variant and T2DM ($p > 0.05$). Therefore, the overall findings suggest that the E224K variant does not contribute to T2DM susceptibility within this dataset.

The absence of the variant is consistent with existing literature indicating that rare variants in monogenic diabetes genes exhibit strong population specificity (Flannick and Florez, 2016; Lian et al., 2023). While the E224K mutation has been previously reported in Indo-Trinidadian and selected Asian populations (Cockburn et al., 2004), its distribution appears limited and not universally observed. Thus, the current findings reinforce the concept that genetic determinants of T2DM vary significantly across populations.

Although the E224K variant has been shown to impair *PDX1* function and insulin gene transcription (Cockburn et al., 2004), its absence in this cohort suggests that alternative genetic mechanisms or environmental factors may play a more prominent role in T2DM pathogenesis in Sri Lanka. This aligns with the understanding that T2DM is a complex polygenic disorder involving multiple low-effect variants and gene–environment interactions (McCarthy, 2017, Mahajan et al., 2018).

PCR amplification and gel electrophoresis results demonstrated successful amplification in 10 out of 12 samples, confirming the effectiveness of the primer design and reaction conditions for most samples. However, the failure to amplify two samples (CC018 and DM019), even after increasing PCR cycles from 35 to 40, highlights important methodological challenges. These failures can be attributed to multiple factors, including DNA degradation due to longterm storage (1 Year) , suboptimal purity, and the presence of PCR inhibitors, as indicated by variability in A260/A280 and A260/A230 ratios.

Importantly, several procedural and environmental factors identified during laboratory work may have contributed to these issues. Initial absence of bands during gel electrophoresis suggested inefficiencies in PCR setup. PCR tubes were occasionally left open for extended periods outside controlled environments, increasing the likelihood of contamination from airborne particles and environmental inhibitors. Labelling and handling were performed outside a laminar flow hood, and delays in reaction setup, prolonged exposure of reagents, and

repeated opening of tubes may have reduced enzyme activity and reaction efficiency. These observations are consistent with evidence that PCR is highly sensitive to contamination and environmental exposure (Lorenz, 2012).

From a troubleshooting perspective, several critical improvements can be proposed based on this experience. Firstly, all PCR setup procedures should be conducted in a dedicated clean area or laminar flow hood to minimise contamination risk. Secondly, strict workflow separation between pre-PCR and post-PCR areas should be maintained to prevent cross-contamination. Thirdly, reagents should be aliquoted into smaller volumes to avoid repeated freeze–thaw cycles and prolonged exposure. Rapid and precise pipetting using calibrated micropipettes is essential to maintain reaction consistency.

In cases of amplification failure, additional troubleshooting strategies should be employed. These include re-extraction of DNA to improve template quality, dilution of DNA samples to reduce inhibitor concentration, optimisation of annealing temperatures through gradient PCR, and adjustment of $MgCl_2$ concentration to enhance enzyme performance. The use of highfidelity or inhibitor-resistant polymerases can further improve amplification success, particularly for low-quality or archived samples. Incorporating internal positive controls would also allow differentiation between true amplification failure and technical error.

Although increasing PCR cycle numbers to 40 improved amplification in some samples, this approach has limitations, as it may increase non-specific amplification and background noise. Therefore, optimisation of reaction conditions is preferable to simply increasing cycle numbers. Sanger sequencing results were consistent and reliable for all successfully amplified samples. Clear single peaks corresponding to guanine at position c.670 confirmed the absence of the E224K mutation. A minor secondary peak observed in one sample (CC023) likely represents background noise or sequencing artefact rather than a true heterozygous mutation. Such artefacts are commonly associated with template quality, dye terminator imbalance, or signal noise (Liu et al., 2021), emphasising the importance of careful chromatogram interpretation.

Statistical analyses further supported the primary findings. Fisher's exact test showed no association between the variant and T2DM, while independent samples t-tests for age and BMI and chi-square analysis for gender distribution also showed no statistically significant differences ($p > 0.05$). These findings indicate that the study groups were broadly comparable, although variability within groups suggests some heterogeneity, which may influence interpretation.

A major limitation of this study is the small sample size, which significantly reduces statistical power. Based on calculations provided in Appendix A and supported by literature (Charan and

Biswas, 2013), approximately 97 participants per group would be required to detect a moderate effect size for a rare variant such as E224K. From a practical perspective, the current study demonstrates that a sample size of 12 is insufficient for detecting rare genetic variants or establishing meaningful associations. Therefore, future studies should aim for a minimum of 100–150 participants per group to achieve adequate statistical power, with larger cohorts required for detecting smaller effect sizes.

Additional limitations include the use of archived DNA samples, which may compromise quality, and technical challenges encountered during laboratory procedures. Furthermore, the investigation was limited to a single variant, whereas T2DM is influenced by multiple genetic and environmental factors.

Despite these limitations, this study provides valuable preliminary data and addresses a significant research gap in the Sri Lankan context. It establishes that the *PDX1* E224K variant is not a common contributor to T2DM in this population and highlights the importance of population-specific genetic studies.

Future research should prioritise large-scale, well-powered studies incorporating diverse populations to validate these findings. Expanding genetic analysis to include multiple variants, full gene sequencing, or next-generation sequencing approaches would provide a more comprehensive understanding of genetic susceptibility. Integration of clinical, biochemical, and functional data will further enhance interpretation.

From a methodological standpoint, future studies should implement stringent laboratory practices, including controlled PCR environments, improved sample handling protocols, and optimisation of amplification conditions. Ensuring high-quality DNA extraction and minimising contamination risks will be critical for generating reliable data.

In summary, this study demonstrates that the *PDX1* E224K variant is absent in the analysed Sri Lankan cohort and does not contribute significantly to T2DM susceptibility within this sample. These findings reinforce the population-specific nature of genetic risk factors and highlight the need for larger, more comprehensive studies.

To further advance this research, future investigations should incorporate increased sample sizes, improved experimental design, and broader genetic analysis. Such approaches will generate more robust and meaningful data, ultimately contributing to improved understanding, diagnosis, and management of T2DM.

Conclusion

This study investigated the prevalence of the PDX1 E224K (c.670G>A) variant and its association with Type 2 Diabetes Mellitus (T2DM) in a Sri Lankan population. The variant was not detected in any of the analysed samples, and no statistically significant association with T2DM was observed. These findings do not support the initial hypothesis and suggest that the E224K variant is either absent or extremely rare in this cohort.

The study objectives were achieved through successful PCR amplification, sequencing, and statistical analysis. The results indicate that this variant does not contribute significantly to T2DM susceptibility in the studied population, supporting the understanding that T2DM is a complex, polygenic disorder influenced by multiple genetic and environmental factors.

Despite limitations such as small sample size, archived DNA quality, and technical challenges during laboratory procedures, this study provides important preliminary data and highlights the need for population-specific genetic research.

In conclusion, the PDX1 E224K variant does not appear to play a major role in T2DM in this Sri Lankan cohort. Future studies with larger sample sizes, improved methodologies, and broader genetic analysis are recommended to generate more robust and comprehensive insights into the genetic basis of T2DM.

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Appendix

Appendix A.-Sample size calculation for detection of *PDX1* E224k variant

Assumptions:

- Allele frequency in controls (p_2) = 0.01
- Expected frequency in cases (p_1) = 0.10
- Significance level (α) = 0.05
- Statistical power = 80%

Formula for Sample Size (Two-Proportion Comparison):

$$n = \frac{(Z_{\alpha/2} + Z_{\beta})^2 \times [p_1(1 - p_1) + p_2(1 - p_2)]}{(p_1 - p_2)^2}$$

Where: $Z_{\alpha/2} = 1.96$ $Z_{\beta} = 0.84$

Calculation:

$$n = \frac{(1.96 + 0.84)^2 \times [(0.10 \times 0.90) + (0.01 \times 0.99)]}{(0.10 - 0.01)^2}$$

$$n = \frac{(2.80)^2 \times (0.09 + 0.0099)}{(0.09)^2}$$

$$n = \frac{7.84 \times 0.0999}{0.0081}$$

$$n \approx 96.6 \approx 97$$

B.- Primers used for amplification of the PDX1 target region, including synthesis and quality details

Primer	Sequence (5'–3')	T _m (°C)	G C (%)	MW (calculated)	MW (measured)	OD	Yield (nmol)	Scale (μmol)	Purification
PDX1 – E224 K–F2	5'–GGAGGAGAACAAGCG GGACG–3'	64.3	63	5960.8	5976.1	7.5	32	0.05	MOPC
PDX1 – E224 K–R2	5'–CCTCTCATCGTGGTTCC TGC–3'	63.9	60	6011	6008.8	5.6	30	0.05	MOPC

C.- Risk Assessment for Laboratory Procedures

Activity / Procedure	Potential Hazard	Risk Level	Possible Consequences	Control Measures	Residual Risk
Handling human DNA samples (biobank samples)	Biological exposure (low risk, anonymised samples)	Low	Minimal risk of contamination or exposure	Use of gloves, lab coat; samples anonymised; follow biosafety guidelines	Very Low
PCR setup (pipetting reagents, primers, DNA)	Chemical exposure (buffers, DMSO), contamination	Low	Skin/eye irritation; sample contamination	Wear PPE (gloves, lab coat); use micropipettes correctly; work in clean area	Very Low
Thermal cycler use	Heat exposure	Low	Minor burns if mishandled	Avoid direct contact with heated lid/block; follow manufacturer instructions	Very Low
Agarose gel preparation	Hot agarose, chemical exposure	Medium	Burns from hot agarose; inhalation of fumes	Allow agarose to cool before handling; wear gloves; work in ventilated area	Low

Appendix

Ethidium bromide staining	Mutagenic chemical	High	DNA damage, toxicity if exposed	Use minimal quantities; wear gloves; dispose as hazardous waste; use safer alternatives if available	Low
Gel electrophoresis (power supply)	Electrical hazard	Medium	Electric shock	Ensure equipment is dry; do not touch buffer during operation; follow SOPs	Low
UV transilluminator use	UV radiation exposure	Medium	Eye/skin damage	Use UV shield; wear protective goggles; minimize exposure time	Low
PCR product purification (ethanol, beads)	Chemical exposure (ethanol)	Low	Irritation, inhalation risk	Use in ventilated area; avoid inhalation; wear PPE	Very Low
Sanger sequencing reagents (BigDye)	Chemical hazard	Low	Mild irritation if exposed	Follow manufacturer safety guidelines; wear PPE	Very Low
Centrifugation	Mechanical hazard	Low	Injury if improperly balanced	Balance tubes properly; close lid before operation	Very Low
Waste disposal (chemical & biological)	Environmental contamination	Medium	Hazardous exposure	Dispose waste according to institutional biosafety protocols	Low

D.- Raw data for Ethical clearance Form

	Research Ethics Committee National Hospital of Sri Lanka, Colombo 10. Tel: +94 11 269 1111 Fax: +94 11 2698443 Email: ddg@nhsl.health.gov.lk
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 Dr. Uditha Bulugahapitiya } Co - Investigators

Dr. Manilka Sumanatilleke - Supervisor

Ref: AAJ/ETH/COM/2024/OCT

Title: "Assessing the prevalence and understanding genetic heterogeneity of people with monogenic diabetes in Sri Lanka- A case-control study at a tertiary care centre".

Thank you for submitting the above **revised** research proposal, which was considered by the Research Ethics Committee for full board review, at its meeting on 24.10.2024.

-This approval is valid for one year, and the committee requires that you furnish it with an annual report on the study's progress beginning on 24.10.2024.

-This approval relates to the ethical content of the study only, and you are responsible for the following:

Negotiating individual arrangements with the Heads of Service departments in those situations where the use of their resources is involved.

Yours Sincerely,


Dr. W.K. Wickremasinghe
 Deputy Director General
 National Hospital of Sri Lanka
 Deputy Director General
 National Hospital of Sri Lanka
 Colombo 10.

Appendix

E.- Participant consent form English
language:

Assessing the prevalence and understanding genetic heterogeneity of people with monogenic diabetes in Sri Lanka-A Case-control study at a tertiary care centre

1. Serial Number:

2. Clinic Number:

3. Age:

4. Gender:

M	F
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5. Address:

6. Contact Number:

7. Ethnicity:

Sinhala	Tamil	Muslim	Other
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8. Occupation:

9. Diabetes Mellitus

Yes	No
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10. Type of DM

Type 1	
Type 2	
MODY	
Not Sure	
Other	

11. Age at presentation:

12. Duration of illness:

13. History of DKA

Yes	No
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14. Number of episodes

Only at diagnosis	
Number of episodes within last year	

15. Comorbidities

Hypertension	
Dyslipidemia	
IHD	
Other	

16. Complications

Nephropathy	
Neuropathy	
Retinopathy	
Macrovascular complications	

17. Syndromic features

Deafness	
Ocular muscle defects	
Retinal changes	
Liver adenomas	
Renal cysts	
Alkalosis	
Hypomagnesemia	
Developmental delay	

18. Family History

Relationship			
Age at diagnosis			
Type of DM			

19. Height:.....

20. Weight:.....

21. BMI:.....

22. HbA1C:.....

23. FBS:.....

24. GAD Antibody:.....

25. C peptide:.....

26. Identified Mutation:.....

27. Treatment

Patient's Informed Consent for blood collection

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I hereby consent to participate in testing for monogenic diabetes using a genetic test. I understand that a blood sample will be obtained from me and/or members of my family. I understand that this specimen will be used for the purpose of attempting to determine if I and members of my family are carriers of the disease gene, or are affected with, or at increased risk to someday be affected with this genetic disease.

PATIENT ACKNOWLEDGEMENT

1. I have provided true and reliable personal information, and fully understand the indication, intended purpose and potential risk of this test.
2. I understand that any data obtained using my blood sample may be used for research purposes, and my personal details will be maintained within the laboratory and will not be published in a way that affects my confidentiality
3. I will indemnify Genelabs Medical (Pvt) Ltd from any liability in connection with this test and its results.
4. The participation in research studies is voluntary and I have the right to refuse, or withdraw without penalty by informing the laboratory verbally or in writing

I have read and understood the purpose, scope and limitations of the test outlined above. My signature below acknowledges my voluntary participation in this test.

Signature:

Date:

TO BE COMPLETED BY INVESTIGATOR

Samples Collected by: _____

Samples Collected Place: _____

Signature

Date Received: __/__/____ Time Received: __: __

Sinhala language:

ශ්‍රී ලංකාවේ දියවැඩියා රෝගීන් අතර එක් ජාත්‍යන්තර විකෘතිතාවයක් නිසා ඇතිවන දියවැඩියාවේ ප්‍රවණතාවය සහ ජාන විෂමතාවය පිළිබඳ තෘතීයික සත්කාර මධ්‍යස්ථානයක සිදුකරන අධ්‍යයනයක්

කැමැත්ත ප්‍රකාශ කිරීමේ පත්‍රිකාව

A කොටස - සහභාගීවන්නන් විසින් පිරවිය යුතුය.

කරුණාකර පහත පෝරමය සම්පූර්ණයෙන් පුරවන්න.

1. 1. ඔබ විසින් තොරතුරු පත්‍රිකාව කියවා තේරුම් ගත්තේද? (එහි පිටපතක් ඔබ ලබා ගන්න.) ඔව්/ නැත
2. 2. ඔබට මෙම පර්යේෂණය පිළිබඳ සාකච්ඡා කිරීමට හා ප්‍රශ්න ඇසීමට අවස්ථාව ලැබුණිද? ඔව්/ නැත
3. 3. අපැහැදිලි තැන් පිළිබඳව ප්‍රශ්න කර නිරවුල් කරගැනීමට අවස්ථාව ලැබුණිද?
4. 4. ඔබ අසන ලද ප්‍රශ්න වලට සතුටුදායක පිළිතුරු ලැබුණිද? ඔව්/ නැත
5. 5. මෙම පර්යේෂණය පිළිබඳ අවශ්‍ය තරම් තොරතුරු ලැබුණිද? ඔව්/ නැත
6. 6. ඔබට මෙම මෙම පර්යේෂණය පිළිබඳ පැහැදිලි කළේ කවුද?
7. 7. මෙම පර්යේෂණය යෙන් කිසිදු හේතු දැක්වීමකින් තොරව, ඔබට ලැබෙන වෛද්‍ය රැකවරණයට ඔව්/ නැත
බාධා නොවී, ඕනෑම අවස්ථාවක මින් ඉවත් විය හැකි බව දන්නේද?
8. 8. මෙම අධ්‍යයනය සම්බන්ධයෙන් පරීක්ෂකයන් විසින් ඔබෙන් ලබා ගන්නා තොරතුරු අනෙකුත් ඔව්/ නැත
පර්යේෂණ සභායකයින් විසින් පරීක්ෂා කළ හැකිය. සියළුම පෞද්ගලික තොරතුරු ඉතා රහසිගත
ලෙස සලකනු ලැබේ. මෙම පර්යේෂණයන්ට ඔබේ වාර්තා වලට ප්‍රවේශ වීමට ඔබ අවසර
ලබා දෙන්නවාද?
9. 9. මෙම තීරණය ගැනීමට ඔබට ප්‍රමාණවත් කාලයක් ලැබුණිද? ඔව්/ නැත
10. 10. ඔබ මෙම පර්යේෂණයට සහභාගී වීමට එකඟ වේ ද? ඔව්/ නැත

අත්සන :

දිනය:

සහභාගීවන්නාගේ සම්පූර්ණ නම:

දෙවන කොටස විමර්ශක/ ප්‍රධාන පර්යේෂක විසින් පිරවිය යුතුය.

මෙම ස්වේච්ඡා පුද්ගලයාට /සහභාගීවන්නාට මම අධ්‍යයනය පැහැදිලි කර දුන් අතර සහභාගී වීමට කැමැත්තෙන් සිටින බව දන්වා ඇත.

පර්යේෂකයාගේ අත්සන:

දිනය:

පර්යේෂකයාගේ සම්පූර්ණ නම:

රෝගියා දැනුවත් වූ බවට වන එකඟතාවය

ශ්‍රී ලංකාවේ දියවැඩියා රෝගීන් අතර එක් ජාත්‍යන්තර විකෘතිතාවයක් නිසා ඇතිවන දියවැඩියාවේ ප්‍රවණතාවය සහ ජාන විෂමතාවය පිළිබඳ තෘතීයික සත්කාර මධ්‍යස්ථානයක සිදුකරන අධ්‍යයනයක්

ජාන පරීක්ෂණයක් භාවිතයෙන් එක් ජාත්‍යන්තර විකෘතිතාවයක් නිසා ඇතිවන දියවැඩියාව පිළිබඳ පර්යේෂණයට සහභාගී වීමට මම මෙයින් එකඟ වෙමි. මාගේ හා මගේ පවුලේ සාමාජිකයන්ගේ රුධිර සාම්පල දෙකක් (3 ml බැගින්) ලබාගන්නා බව මට වැටහේ. මාගේ මගේ පවුලේ සාමාජිකයන් මෙම රෝගී ජානයේ වාහකයන්ද, එසේත් නැතිනම් මෙම ප්‍රවේණික රෝගය සමඟ යම් දිනක පීඩාවට පත්වීමේ වැඩි අවදානමක් තිබේද යන්න තීරණය කිරීමට උත්සාහ කිරීමට මෙම නිදර්ශනය භාවිතා කරන බව මට වැටහේ.

රෝගියාගේ එකඟතාවය

මම සත්‍ය සහ විශ්වාසදායක පුද්ගලික තොරතුරු සපයා ඇති අතර මෙම පරීක්ෂණයේ ඇගවීම්, අපේක්ෂිත අරමුණු සහ විය හැකි අවදානම් සම්පූර්ණයෙන් අවබෝධ කරගෙන ඇත්තෙමි. මගේ රුධිර සාම්පල භාවිතයෙන් ලබා ගන්නා ඕනෑම දත්තයක් පර්යේෂණ කටයුත්තක් සඳහා භාවිත කළ හැකි බවත් මගේ පෞද්ගලික තොරතුරු රසායනාගාරය තුළ පමණක් පවත්වාගෙන යනු ඇති බවත් මාගේ පෞද්ගලිකත්වයට බලපාන අයුරින් ප්‍රකාශයට පත් නොකරන බවත් මට වැටහේ.

මෙම පරීක්ෂණය සහ ප්‍රතිඵල සම්බන්ධව මම Genelabs Medical (Pvt) Ltd ආයතනය නීති වගකීම් වලින් මුදා හරිමි.

පර්යේෂණ අධ්‍යයනවලට සහභාගීවීම ස්වේච්ඡාවෙන් සිදුවන අතර වාචිකව හෝ ලිඛිතව රසායනාගාරයට දැනුවත්කර මෙම පර්යේෂණයෙන් ඉවත් වීමට මට අයිතියක් ඇත.

ඉහත දක්වා ඇති පර්යේෂණයේ අරමුණ, විෂය පථය සහ සීමාවන් මම නියවා තේරුම් ගෙන ඇත. පහත මගේ අත්සන යෙදීමෙන් මෙම පරීක්ෂණයට මගේ ස්වේච්ඡාව සහභාගීත්වය පිළිගනිමි.

සම්පූර්ණ නම:

අත්සන:

දිනය:

විමර්ශකයා විසින් සම්පූර්ණ කළ යුතුය

සාම්පල එකතු කරන අයගේ නම සහ තනතුර:

සාම්පල එකතු කළ ස්ථානය:

ලැබුණු දිනය:

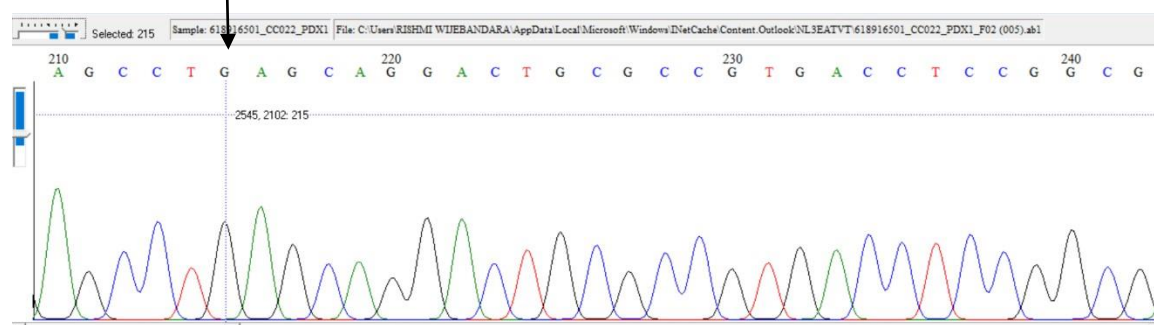
ලැබුණු වෙලාව:

අත්සන:

Appendix F.- Sequencing Results

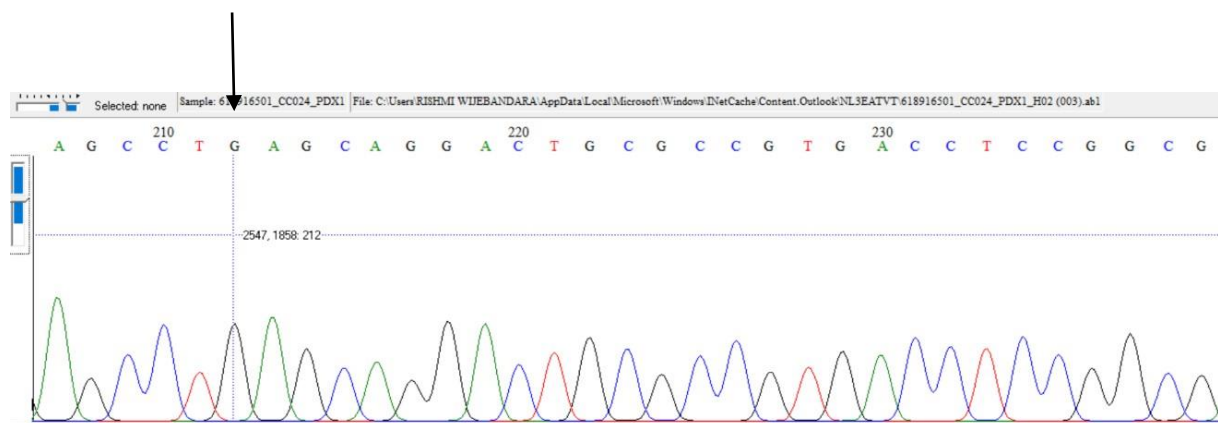
Sample ID- CC022

G at position 670



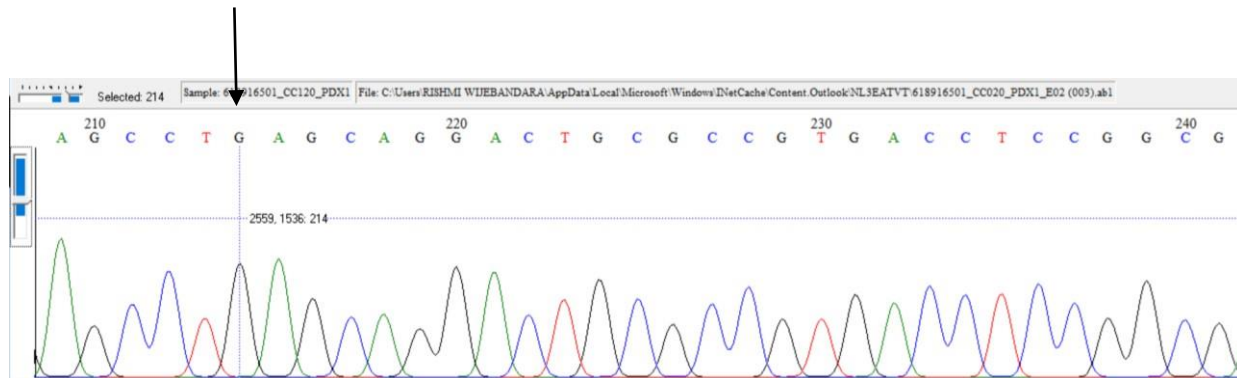
Sample ID- CC024

G at position 670



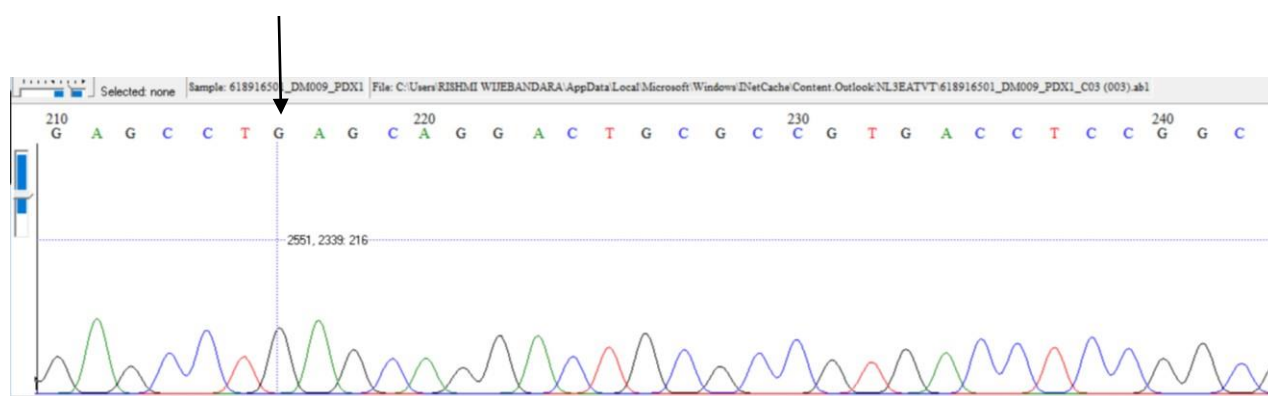
Sample ID- CC020

G at position 670



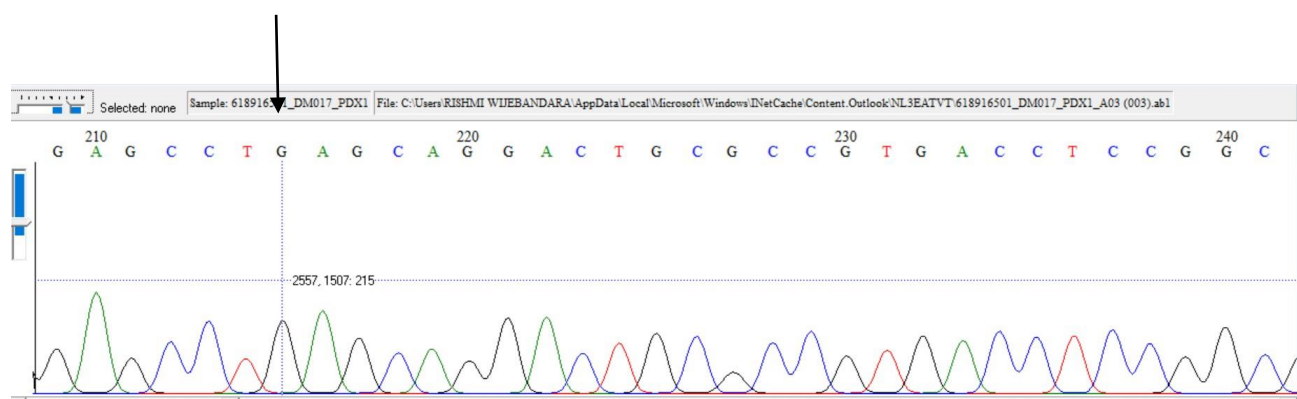
Sample ID- DM009

G at position 670



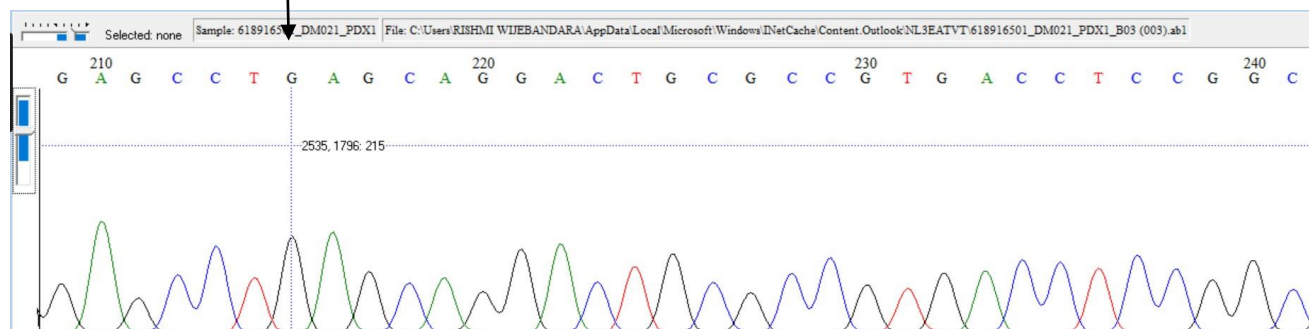
Sample ID- DM017

G at position 670



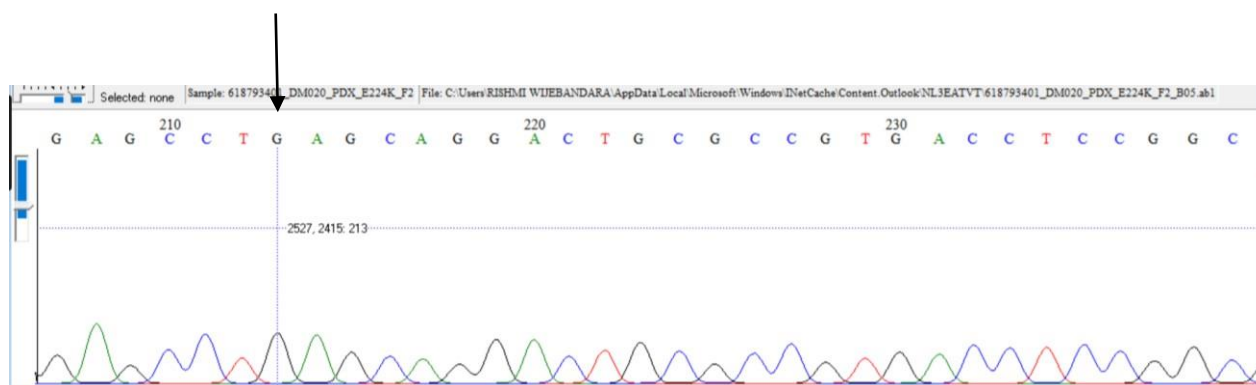
Sample ID- DM021

G at position 670



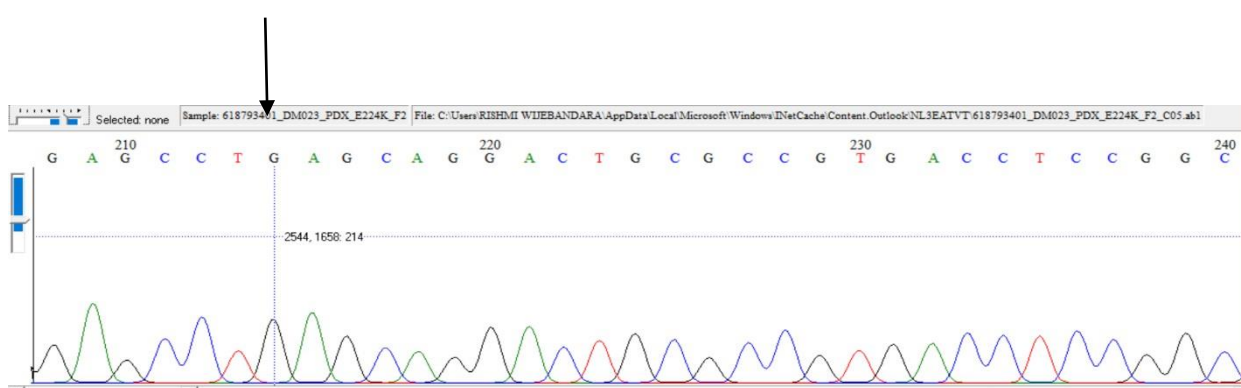
Sample ID- DM020

G at position 670



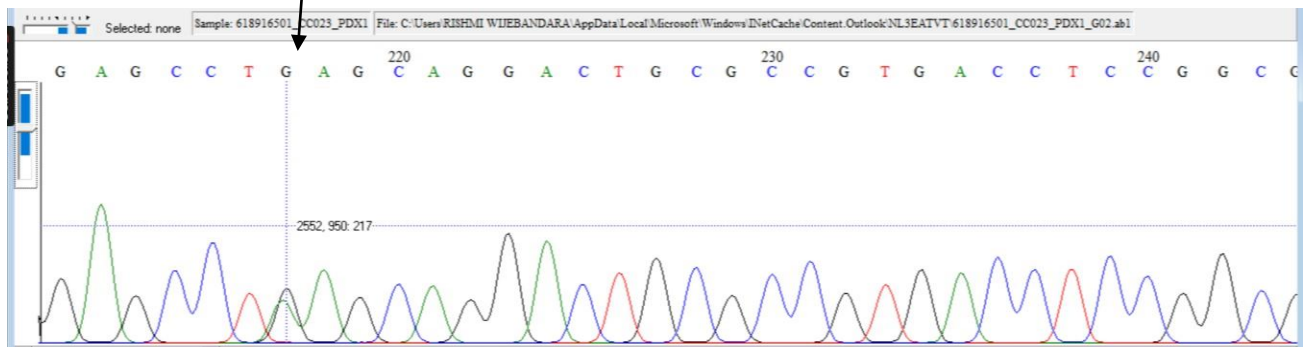
Sample ID- DM023

G at position 670



Sample ID- CC023

G at position 670-Although minor background signal was observed near the peak, the chromatogram showed a single dominant guanine (G) peak at position c.670 without overlapping peaks of comparable intensity, confirming the absence of a heterozygous or homozygous mutation.



Appendix G.- Description of Study Population (Descriptive Statistics)

Mean & SD (Age, BMI)

Case Processing Summary							
	Group	Cases					
		Valid		Missing		Total	
		N	Percent	N	Percent	N	Percent
Age (Years)	Control	6	100.0%	0	0.0%	6	100.0%
	Diabetic	6	100.0%	0	0.0%	6	100.0%
BMI	Control	6	100.0%	0	0.0%	6	100.0%
	Diabetic	6	100.0%	0	0.0%	6	100.0%

Descriptives ^{a,b}				
	Group		Statistic	Std. Error
Age (Years)	Control	Mean	56.00	2.898
		95% Confidence Interval for Mean	Lower Bound	48.55
			Upper Bound	63.45
		5% Trimmed Mean		56.39

		Median		58.50	
		Variance		50.400	
		Std. Deviation		7.099	
		Minimum		43	
		Maximum		62	
		Range		19	
		Interquartile Range		11	
		Skewness		-1.549	.845
		Kurtosis		2.199	1.741
	Diabetic	Mean		61.67	5.432
		95% Confidence Interval for Mean	Lower Bound	47.70	
			Upper Bound	75.63	
		5% Trimmed Mean		61.69	
		Median		61.00	
		Variance		177.067	
		Std. Deviation		13.307	
		Minimum		41	

		Maximum		82	
		Range		41	
		Interquartile Range		17	
		Skewness		-.041	.845
		Kurtosis		1.662	1.741
BMI	Control	Mean		27.6799433850	3.57057072923
		95% Confidence Interval for Mean	Lower Bound	18.5014991257	
			Upper Bound	36.8583876443	
		5% Trimmed Mean		27.4256874128	
		Median		25.8820401850	

		Variance		76.494	
		Std. Deviation		8.7460763771 4	
		Minimum		17.31301939	
		Maximum		42.62347488	
		Range		25.31045549	
		Interquartile Range		12.40118432	
		Skewness		.941	.845
		Kurtosis		1.206	1.741
	Diabetic	Mean		24.417682270 0	2.0725053002 2
		95% Confidence Interval for Mean	Lower Bound	19.090137791 0	
			Upper Bound	29.745226749 0	
		5% Trimmed Mean		24.399290426 1	
		Median		23.908999420 0	
		Variance		25.772	
		Std. Deviation		5.0765804747 5	
		Minimum		17.19178983	
		Maximum		31.97462790	
		Range		14.78283807	
		Interquartile Range		8.20376451	
				Skewness	
Kurtosis				.247	1.741
a. There are no valid cases for Age (Years) when Group = .000. Statistics cannot be computed for this level.					
b. There are no valid cases for BMI when Group = .000. Statistics cannot be computed for this level.					

Age (Years)

Stem-and-Leaf Plots

Age (Years) Stem-and-Leaf Plot for
Group= Control

Frequency Stem & Leaf

1.00 4 . 3
3.00 5 . 389
2.00 6 . 12

Stem width: 10

Each leaf: 1 case(s)

Age (Years) Stem-and-Leaf Plot for
Group= Diabetic

Frequency Stem & Leaf

1.00 Extremes (= <41)
1.00 5 . 8
2.00 6 . 02
1.00 6 . 7
1.00 Extremes (>=82)

Stem width: 10

Each leaf: 1 case(s)

BMI

Stem-and-Leaf Plots

BMI Stem-and-Leaf Plot for
Group= Control

Frequency Stem & Leaf

```

1.00    1 . 7
3.00    2 . 338
1.00    3 . 1
1.00    4 . 2

```

Stem width: 10.00000
Each leaf: 1 case(s)

BMI Stem-and-Leaf Plot for
Group= Diabetic

Frequency Stem & Leaf

```

1.00    1 . 7
4.00    2 . 1347
1.00    3 . 1

```

Stem width: 10.00000
Each leaf: 1 case(s) **Gender %**

Case Processing Summary						
	Cases					
	Valid		Missing		Total	
	N	Percent	N	Percent	N	Percent
Group * Gender	13	100.0%	0	0.0%	13	100.0%

Group * Gender Crosstabulation						
			Gender			Total
				Female	Male	
Group		Count	1	0	0	1
		% within Group	100.0%	0.0%	0.0%	100.0%
	Control	Count	0	4	2	6
		% within Group	0.0%	66.7%	33.3%	100.0%
	Diabetic	Count	0	5	1	6
		% within Group	0.0%	83.3%	16.7%	100.0%

Total	Count	1	9	3	13
	% within Group	7.7%	69.2%	23.1%	100.0%

Appendix H. - Association Between PDX1 E224K Variant and T2DM

Crosstabs

Case Processing Summary						
	Cases					
	Valid		Missing		Total	
	N	Percent	N	Percent	N	Percent
Group * Mutation_Present	13	100.0%	0	0.0%	13	100.0%

Chi-Square Tests				
	Value	df	Asymptotic Significance (2-sided)	Exact Sig. (2-sided)

Pearson Chi-Square	13.000 ^a	4	.011	.077
Likelihood Ratio	7.051	4	.133	.231
Fisher-Freeman-Halton Exact Test	6.352			.231
N of Valid Cases	13			
a. 9 cells (100.0%) have expected count less than 5. The minimum expected count is .08.				

Appendix I. -

T-Test

Independent Samples t-test

Variable	Assumption	F	Sig.	t	df	Sig. (2tailed)	Mean Difference	Std. Error Difference	95% CI Lower	95% CI Upper
Age (Years)	Equal variances assumed	0.646	0.440	0.920	10	0.379	5.667	6.157	-8.052	19.386
	Equal variances not assumed			0.920	7.633	0.386	5.667	6.157	-8.652	19.985

Independent Samples Effect Sizes					
		Standardized ^a	Point Estimate	95% Confidence Interval	
				Lower	Upper
Age (Years)	Cohen's d	10.665	.531	-.636	1.673
	Hedges' correction	11.557	.490	-.587	1.544
	Glass's delta	7.099	.798	-.464	1.995
a. The denominator used in estimating the effect sizes. Cohen's d uses the pooled standard deviation. Hedges' correction uses the pooled standard deviation, plus a correction factor. Glass's delta uses the sample standard deviation of the control group.					

Appendix J.- Comparison of BMI Between Groups

	Levene's Test for Equality of Variances		t-test for Equality of Means					
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	F	Sig.	t	df	Sig. (2tailed)	Mean Difference	Std. Error Difference	95% CI (Lower – Upper)
Equal variance s assumed	1.189	0.301	-0.790	10	0.448	- 3.2622611 2	4.12846866 9	- 12.4610626 to 5.93554032 6
Equal variance s not assumed			-0.790	8.026	0.452	- 3.2622611 2	4.12846866 9	- 12.7772284 to 6.25270621 4

Independent Samples Effect Sizes					
		Standardize r ^a	Point Estimate	95% Confidence Interval	
				Lower	Upper
BM I	Cohen's d	7.15071749 237	-.456	-1.594	.703
	Hedges' correction	7.74939206 425	-.421	-1.471	.649
	Glass's delta	8.74607637 714	-.373	-1.509	.798
a. The denominator used in estimating the effect sizes. Cohen's d uses the pooled standard deviation. Hedges' correction uses the pooled standard deviation, plus a correction factor. Glass's delta uses the sample standard deviation of the control group.					

Appendix K. - Gender Distribution Analysis

Chi-Square Test

Case Processing Summary			
	Cases		
	Valid	Missing	Total

	N	Percent	N	Percent	N	Percent
Group * Gender	13	100.0 %	0	0.0%	13	100.0 %

Chi-Square Tests			
	Value	df	Asymptotic Significance (2-sided)
Pearson ChiSquare	13.481 ^a	4	.009
Likelihood Ratio	7.502	4	.112
N of Valid Cases	13		
a. 9 cells (100.0%) have expected count less than 5. The minimum expected count is .08.			

Appendix L: Materials Used in the Study

Category	Material / Reagent / Equipment	Specification / Description	Manufacturer / Source	Application
Biological Samples	Human genomic DNA	Pre-extracted from peripheral blood	Genelabs Medical Laboratory, Sri Lanka	PCR amplification
Primers	PDX1–E224K–F2	5′–GGAGGAGAACAAGCGG GACG–3′	Commercial synthesis (Genelabs)	Forward primer for PCR
	PDX1–E224K–R2	5′–CCTCTCATCGTGGTTCC TGC–3′	Commercial synthesis (Genelabs)	Reverse primer for PCR
PCR Reagents	5× PCR Master Mix	Includes Taq polymerase, dNTPs, buffer	(Specify brand if known)	DNA amplification
	DMSO	Molecular biology grade	(Specify brand)	Enhances PCR specificity
	Nuclease-free water	Sterile	(Specify brand)	Reaction preparation
Electrophoresis Reagents	Agarose	Molecular biology grade	(Specify brand)	Gel preparation
	1× TAE Buffer	Tris-acetate-EDTA buffer	(Specify brand)	Electrophoresis running buffer
	Ethidium bromide	0.6 µg/mL	(Specify brand)	DNA staining
	DNA ladder	100 bp ladder	Thermo Fisher Scientific, USA	Size reference
Purification Reagents	Magnetic beads	PCR purification beads	(Specify brand)	PCR cleanup
	Ethanol (80% & 100%)	Analytical grade	(Specify brand)	Washing and purification
Sequencing Reagents	BigDye® Terminator v3.1 Kit	Cycle sequencing kit	Applied Biosystems, USA	Sanger sequencing
	EDTA	Molecular grade	(Specify brand)	Postsequencing purification

Equipment	Thermal Cycler	Veriti™ Thermal Cycler	Applied Biosystems, USA	PCR amplification
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	Gel electrophoresis unit	Standard horizontal gel system	Bio-Rad (or specify)	DNA separation
	Gel documentation system	Gel Doc™ XR+	Bio-Rad, USA	Visualization
	Spectrophotometer	NanoDrop	Thermo Fisher Scientific, USA	DNA quantification
	Genetic Analyzer	SeqStudio™	Applied Biosystems, USA	Sequence analysis
	Centrifuge	Bench-top centrifuge	(Specify brand)	Sample processing
Software	BioEdit v7.0.9	Sequence editing software	Freeware	Chromatogram analysis
	ClustalW	Alignment tool	Integrated software	Sequence alignment
	IBM SPSS v27	Statistical software	IBM Corp., USA	Data analysis

Appendix M. – XL sheet with sample details

	A	B	C	D	E	F	G	H	I	J	K	
1	Sample_ID	Group	Diabetes_Status	Mutation_Present	Gender	Age (Years)	Ethnic	Diabetic_Duration (Years)	BMI	Specimen	DNA concentration-Spectrophotometer(ng/ul)	Other Control
2	DM009	Diabetic	T2DM	No	Female	60	Sinhala	3	27.76620852	Blood		51.3 35 Cycle
3	DM017	Diabetic	T2DM	No	Female	67	Sinhala	8	21.75546853	Blood		54.95 35 cycle
4	DM020	Diabetic	T2DM	No	Male	82	Sinhala	5	24.50894577	Blood		60.85 40 cycle
5	DM021	Diabetic	T2DM	No	Female	62	Sinhala	17	31.9746279	Blood		55.15 40 cycle
6	DM023	Diabetic	T2DM	No	Female	58	Sinhala	20	17.19178983	Blood		31.65 40 cycle
7	DM019	Diabetic	T2DM	N/A	Female	41	Tamil	7	23.30905307	Blood		32.35 No Bands
8	CC016	Control	Healthy	No	Male	62	Muslim	No	28.32658037	Blood		107.95 40 cycles
9	CC020	Control	Healthy	No	Male	43	Tamil	No	23.14049587	Blood		211.35 40 cycles
10	CC022	Control	Healthy	No	Female	58	Sinhala	No	17.31301939	Blood		91.1 40 cycles
11	CC023	Control	Healthy	No	Female	53	Tamil	No	42.62347488	Blood		96.85 35 cycle
12	CC024	Control	Healthy	No	Female	61	Sinhala	No	23.4375	Blood		174.9 40 cycles
13	CC018	Control	Healthy	N/A	Female	59	Muslim	No	31.2385898	Blood		2.65 No bands
14												
15												